

Benchmark results compiled with -O2: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	158.139	156.777	1244.762	4.739	145.858	296.705	262.658	4.877	87.977	188.524	1041.686	4.958	4.859	–	–	–
countdown.deep.10000	1277.935	574.768	2282.623	2747	148.663	283.878	164.357	5.032	169.263	193.371	3568.728	4.997	4.979	65.663	–	–
catch.shallow.10000	272.479	36.039	1128.573	20.646	1069.168	376285.4	249.796	16.313	74.952	846.243	847.725	20.833	20.63	–	–	–
catch.deep.10000	3247.017	38.622	6597.4	1683.513	1069.771	404932.2	178.258	16.52	239.206	850.936	850.293	20.71	20.975	167.221	–	–
local.shallow.10000	930.697	456.232	2147.605	5.046	547.715	404539.6	201.55	2.536	140.81	795.92	792.197	5.041	5.035	–	–	–
local.deep.10000	3825.736	459.271	5074.791	5.043	552.473	300001.3	204.35	2.535	219.692	799.043	805.04	5.045	5.032	206.336	–	–
nondet.shallow.32	1944.997	3295.872	7559.541	255.333	–	24826.175	769.096	80.617	894.248	1789.741	1793.108	224.91	243.832	–	359.07	360.041
nondet.deep.32	5442.916	5585.028	7708.309	2452.9	–	50519.675	769.511	80.799	616.905	5776.244	5770.919	252.748	244.1	1006.458	63032.675	63848.85

Benchmark results compiled with -O2: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	3337.3%	3308.5%	26268.6%	100.0%	3078.1%	6261.5%	5543.0%	102.9%	1856.6%	3978.5%	21983.0%	104.6%	102.5%	–	–	–
countdown.deep.10000	25608.2%	11544.6%	45848.0%	55175.3%	2986.0%	5701.9%	3301.2%	101.1%	3399.7%	3884.0%	71680.3%	100.4%	100.0%	1318.9%	–	–
catch.shallow.10000	1670.4%	220.9%	6918.4%	126.6%	6554.3%	2306723.6%	1531.3%	100.0%	459.5%	5187.7%	5196.8%	127.7%	126.5%	–	–	–
catch.deep.10000	19655.0%	233.8%	39935.8%	10190.7%	6475.6%	2451160.6%	1079.0%	100.0%	1448.0%	5150.9%	5147.0%	125.4%	127.0%	1012.2%	–	–
local.shallow.10000	36694.6%	17987.9%	84673.6%	198.9%	21594.7%	15949776.3%	7946.5%	100.0%	5551.7%	31380.7%	31233.9%	198.8%	198.5%	–	–	–
local.deep.10000	150917.6%	18117.3%	200190.2%	199.0%	21793.9%	15384763.8%	8061.2%	100.0%	8666.4%	31520.6%	31757.2%	199.0%	198.5%	8139.5%	–	–
nondet.shallow.32	2412.6%	4088.3%	9377.1%	316.7%	–	30795.2%	954.0%	100.0%	1109.3%	2220.1%	2224.2%	279.0%	302.5%	–	445.4%	446.6%
nondet.deep.32	6736.4%	6912.3%	9540.2%	3035.8%	–	62525.4%	952.4%	100.0%	763.5%	7148.9%	7142.4%	312.8%	302.1%	1245.6%	78012.1%	79022.2%

Benchmark results compiled with -O2: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	17601.51	16001.51	13361526	24	1121559	2087371	2480238	24	1040119	3441159	14241107	24	24	–	–	–
countdown.deep.10000	17760454	10400166	37363350	27121228	1136503	2088076	2000190	24	3920263	3442679	41443886	24	24	1200099	–	–
catch.shallow.10000	4069590	65593	7764882	65585	7846338	413419837	1687791	65584	723951	6081379	6081379	65584	65585	–	–	–
catch.deep.10000	16870417	66059	55153624	16165078	7864083	414304480	1527783	65584	3367871	6083939	6083939	65585	65585	1527782	–	–
local.shallow.10000	7205046	4793270	12629902	24	4326213	409639753	1669447	24	1029423	5513566	5513563	24	24	–	–	–
local.deep.10000	16006005	4793725	30232179	24	4341356	410400075	1589447	24	2549551	5515811	5515812	64	64	1589471	–	–
nondet.shallow.32	13454095	15889914	43492505	3600903	–	111260655	1531315	108343	1532750	22646046	22646045	1987326	2003982	–	3664086	3664078
nondet.deep.32	67560272	45645780	43569846	26810526	–	383226312	1531395	108343	1535147	68636406	68636435	1969094	2010671	7408468	469818611	469818611

Benchmark results compiled with -O2: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	7333962.5%	6667295.8%	55673025.0%	100.0%	4673162.5%	8697379.2%	10334325.0%	100.0%	4333829.2%	14338162.5%	59337945.8%	100.0%	100.0%	–	–	–
countdown.deep.10000	74801891.7%	43334025.0%	15568625.0%	113005116.7%	4735429.2%	8700316.7%	8334125.0%	100.0%	16334429.2%	14344495.8%	172682858.3%	100.0%	100.0%	5000412.5%	–	–
catch.shallow.10000	6205.2%	100.0%	11839.6%	100.0%	11963.8%	63066.9%	2573.5%	100.0%	1103.9%	9272.7%	9272.7%	100.0%	100.0%	–	–	–
catch.deep.10000	25723.4%	100.7%	84096.2%	24647.9%	11990.9%	631715.8%	2329.5%	100.0%	5135.2%	9276.6%	9276.6%	100.0%	100.0%	2329.5%	–	–
local.shallow.10000	30021025.0%	19971958.3%	52624591.7%	100.0%	18025887.5%	1706832304.2%	6956029.2%	100.0%	4289262.5%	22973191.7%	22973179.2%	100.0%	100.0%	–	–	–
local.deep.10000	66691687.5%	19973854.2%	125967412.5%	100.0%	18088983.3%	1710000312.5%	6622791.7%	100.0%	10623129.2%	22982545.8%	22982550.0%	266.7%	266.7%	6622795.8%	–	–
nondet.shallow.32	12418.1%	14666.3%	40143.3%	3329.2%	–	102693.0%	1413.4%	100.0%	1414.7%	20902.2%	20902.2%	1834.3%	1849.7%	–	3381.9%	3381.9%
nondet.deep.32	62357.8%	42130.8%	40214.7%	24746.0%	–	353715.8%	1413.5%	100.0%	1416.9%	63351.0%	63351.1%	1817.5%	1855.8%	6838.0%	433640.0%	433640.0%

Benchmark results compiled with -O0: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	789.915	523.344	1747.473	1680.941	719.506	883.371	1616.242	172.57	1631.315	849.623	2196.878	651.925	569.925	–	–	–
countdown.deep.10000	2944.506	1023.612	2840.553	10658.356	730.042	884.606	1693.118	172.819	3241.595	1425.188	5197.341	5190.506	6013.341	1712.87	–	–
catch.shallow.10000	1088.659	308.105	2790.247	3350.238	2142.367	376211.9	944.955	349.029	1065.443	1633.452	1639.996	254.99	226.36	–	–	–
catch.deep.10000	815.728	317.738	7000.387	15921.369	2161.777	389295.3	1022.406	367.071	2956.667	1657.329	1634.073	3924.427	4054.966	1126.093	–	–
local.shallow.10000	4537.391	672.329	2644.162	1408.634	1066.513	403942.6	621.008	297.651	647.735	1387.716	1384.28	351.432	326.969	–	–	–
local.deep.10000	13426.506	673.914	5601.697	5921.959	1074.391	390857.5	625.165	298.518	1620.895	1386.891	1380.403	817.977	1606.246	651.768	–	–
nondet.shallow.32	7541.894	6065.397	13174.044	2144.384	–	28236.275	4136.884	5796.009	4190.194	2998.684	3029.122	2459.477	1975.917	–	7316.259	6822.278
nondet.deep.32	15624.538	8130.309	22502.537	10549.644	–	55939.475	4273.619	5636.734	9617.172	7717.512	7705.544	6761.516	7237.45	4577.277	80142.25	80554.825

Benchmark results compiled with -O0: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	457.7%	303.3%	1012.6%	974.1%	416.9%	511.9%	936.6%	100.0%	945.3%	492.3%	1273.0%	377.8%	330.3%	–	–	–
countdown.deep.10000	1703.8%	592.3%	1643.7%	6167.4%	422.4%	511.9%	979.7%	100.0%	1875.7%	824.7%	3007.4%	3003.4%	3479.6%	991.1%	–	–
catch.shallow.10000	480.9%	136.1%	1232.7%	1480.0%	946.4%	166200.9%	417.5%	154.2%	470.7%	721.6%	724.5%	112.6%	100.0%	–	–	–
catch.deep.10000	2554.2%	100.0%	2203.2%	5010.8%	680.4%	122520.7%	321.8%	115.5%	930.5%	521.6%	514.3%	1235.1%	1276.2%	354.4%	–	–
local.shallow.10000	1524.4%	225.9%	888.3%	473.2%	358.3%	135710.0%	208.6%	100.0%	217.6%	466.2%	465.1%	118.1%	109.8%	–	–	–
local.deep.10000	4497.7%	225.8%	1876.5%	1983.8%	359.9%	130932.8%	209.4%	100.0%	543.0%	464.6%	462.4%	274.0%	538.1%	218.3%	–	–
nondet.shallow.32	381.7%	307.0%	666.7%	108.5%	–	1429.0%	209.4%	100.0%	212.1%	151.8%	153.3%	124.5%	100.0%	–	370.3%	345.3%
nondet.deep.32	365.6%	190.2%	526.5%	246.9%	–	1308.9%	100.0%	131.9%	225.0%	180.6%	180.3%	158.2%	169.4%	107.1%	1875.3%	1884.9%

Benchmark results compiled with -O0: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	10640932	6080470	17281854	11920939	5202067	6268901	10801414	720158	11601190	7041755	19361708	4080259	2880411	-	-	-
countdown.deep.10000	38643721	14881331	41284538	63365225	5218156	6269554	11213142	720158	15371608	9444807	48965510	30882788	30882856	11365109	-	-
catch.shallow.10000	7913898	2571781	22551529	13075276	12013609	415980479	5433993	1462327	6393803	8161740	8161738	1673342	1593382	-	-	-
catch.deep.10000	33116291	2572597	57554506	64806104	12032261	416787571	5845708	1462327	10161449	8165034	8164589	16926382	18046463	5837709	-	-
local.shallow.10000	14981594	5771835	14008665	7688309	6726756	411560053	4008470	1767853	4648323	7193845	7193848	2327805	2327801	-	-	-
local.deep.10000	34184872	5772653	31611425	28499046	6742821	412321543	4020190	1767858	6818550	7197133	7196629	5048099	9048614	4012067	-	-
nondet.shallow.32	53827085	34532145	81643812	18238382	-	131920581	24814459	53824600	29682954	30881964	30881963	17396977	12390837	-	74123065	69116580
nondet.deep.32	137695451	64288376	200681296	63405435	-	403887165	26178259	53824598	42407659	80929174	80928672	43362122	39124538	28604223	54361446	53910278

Benchmark results compiled with -O0: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.lstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.listt	handrolled.listt
countdown.shallow.10000	1477.6%	844.3%	2399.7%	1655.3%	722.4%	870.5%	1499.9%	100.0%	1610.9%	977.8%	2688.5%	566.6%	400.0%	—	—	—
countdown.deep.10000	5366.0%	2066.4%	5732.7%	8798.8%	724.6%	870.6%	1557.0%	100.0%	2134.5%	1311.5%	6799.3%	4288.3%	4288.3%	1578.1%	—	—
catch.shallow.10000	541.2%	175.9%	1528.5%	894.1%	821.5%	28446.5%	371.6%	100.0%	437.2%	558.1%	558.1%	114.4%	109.0%	—	—	—
catch.deep.10000	2264.6%	175.9%	3935.8%	4431.7%	822.8%	28501.7%	399.8%	100.0%	695.1%	558.4%	558.3%	1157.5%	1234.1%	399.2%	—	—
local.shallow.10000	847.4%	326.5%	792.4%	434.9%	380.5%	23280.2%	226.7%	100.0%	262.9%	406.9%	406.9%	131.7%	131.7%	—	—	—
local.deep.10000	1933.7%	326.5%	1788.1%	1611.6%	381.4%	23323.5%	227.4%	100.0%	385.7%	407.1%	407.1%	285.6%	511.8%	226.9%	—	—
nondet.shallow.32	434.4%	278.7%	658.9%	147.2%	—	1064.7%	200.3%	434.4%	239.6%	249.2%	249.2%	140.4%	100.0%	—	598.2%	557.8%
nondet.deep.32	526.0%	245.6%	766.6%	242.2%	—	1542.8%	100.0%	205.6%	162.0%	309.1%	309.1%	166.7%	149.5%	109.3%	2076.6%	2059.4%